

Web Application Development Practical - Node.js Static Website

Practical: Create a Node.js Application which serves a static website.

Aim: To create and run a Node.js application that serves a responsive static website using Express framework.

Requirements:

1. Node.js Installed
2. VS Code or any Editor
3. Internet Browser
4. Express Package

Step 1: Install Node.js from official website.

Step 2: Create a project folder named staticwebsite.

Step 3: Open terminal inside the project folder.

Step 4: Run command `npm init -y` to initialize Node project.

Step 5: Install Express using command `npm install express`.

Step 6: Create folders named public, css and images.

Step 7: Create index.html inside public folder.

Step 8: Create style.css file inside css folder.

Step 9: Create server.js file in root directory.

Step 10: Write Node.js server code using Express.

Step 11: Run the application using `node server.js` command.

Step 12: Open browser and type `http://localhost:3000`.

Folder Structure

```
staticwebsite/  
├──  
│   ├── server.js  
│   ├── package.json  
│   ├──  
│   ├── public/  
│   │   └── index.html  
│   ├──  
│   ├── css/  
│   │   └── style.css
```

server.js

```
const express = require('express');  
const path = require('path');  
  
const app = express();  
  
app.use('/css', express.static(path.join(__dirname, 'css')));
```

```

app.get('/', (req, res) => {
  res.sendFile(path.join(__dirname, 'public', 'index.html'));
});

const PORT = 3000;

app.listen(PORT, () => {
  console.log(`Server running on http://localhost:${PORT}`);
});

```

index.html

```

<!DOCTYPE html>
<html>
<head>
  <title>Node.js Static Website</title>
  <link rel="stylesheet" href="/css/style.css">
</head>
<body>

  <header>
    <h1>Welcome to Node.js Website</h1>
    <p>Static Website using Node.js and Express</p>
  </header>

  <section class="container">

    <div class="card">
      <h2>Fast Performance</h2>
      <p>Node.js provides fast and scalable applications.</p>
    </div>

    <div class="card">
      <h2>Easy Routing</h2>
      <p>Express framework simplifies routing process.</p>
    </div>

    <div class="card">
      <h2>Responsive Design</h2>
      <p>The website works properly on all devices.</p>
    </div>

  </section>

</body>
</html>

```

style.css

```

body{
  margin:0;
  font-family:Arial;
  background:#f4f4f4;
}

header{
  background:#343a40;
  color:white;
  padding:30px;
  text-align:center;
}

.container{
  display:flex;
  justify-content:center;
  gap:20px;
  padding:30px;
  flex-wrap:wrap;
}

.card{
  background:white;
  width:250px;
  padding:20px;
  border-radius:10px;
}

```

```
        box-shadow:0px 2px 8px gray;
        text-align:center;
    }

    .card h2{
        color:#007bff;
    }
```

Commands to Run Application

```
npm init -y
npm install express
node server.js
```

Explanation:

Node.js is a JavaScript runtime used for server-side development. Express framework is used to simplify server creation and routing. The server.js file creates the server and serves static files like HTML and CSS. The HTML file creates webpage structure while CSS provides styling and responsiveness.

Result: The Node.js application was successfully created and the static responsive website was served in browser.